Climate Predictability in the Context of Global Warming: Preliminary Numerical Experiments

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1 Introduction

The predictability of climate on decadal timescales (i.e., timescales of a few years to a few decades) is not well understood. By this statement we mean the following. On the intraseasonal timescale, there is evidence that numerical forecasts can be made that improve on climatology (e.g., Shukla *et al.* 2000). Much of the source of this skill arises from predicting the conditions of the tropical ocean, and the effects this has on the global atmosphere. Forecasts at this timescale are in some ways well-posed initial-value problems. In some contrast, at the multi-decadal timescale, the response to greenhouse gases may dominate that of natural variability. That is to say, the difference between the climate of a century hence (given typical business-as-usual projections of the increase in greenhouse gas and aerosol concentrations) and that of today is likely to be much larger than the natural variability of climate itself, because the boundary/forcing conditions of the atmosphere have changed so much. Thus, if ones prior estimate is simply the climatology of today, one might say that predictability also resides at the multi-decadal–centennial timescales.

In contrast to both of these limits, on the decadal timescale the natural variability is likely to be roughly comparable with the forced change in climate over that time-period. Unless some aspects of this variability can be predicted, the useful predictability at this timescale may be limited. This situation is not unique to climate; an analogous situation exists in weather forecasting. Here, forecasts of a few days depend almost entirely on the initial conditions, and can be quite skillful. At the seasonal timescale (which we are now regarding as the long-time limit of weather forecasting), skill emerges by way of the influence of the slowly evolving sea-surface temperature, which provides a boundary condition on the atmosphere. But at the intermediate timescale useful skill is harder to define. Thus, for both climate and weather forecasts, the timescale that is long enough so that the memory of the initial conditions is partially lost, but not long enough for the slowly varying boundary conditions to have impressed themselves on the system, may provide the greatest challenge. It is this timescale we explore in this note.

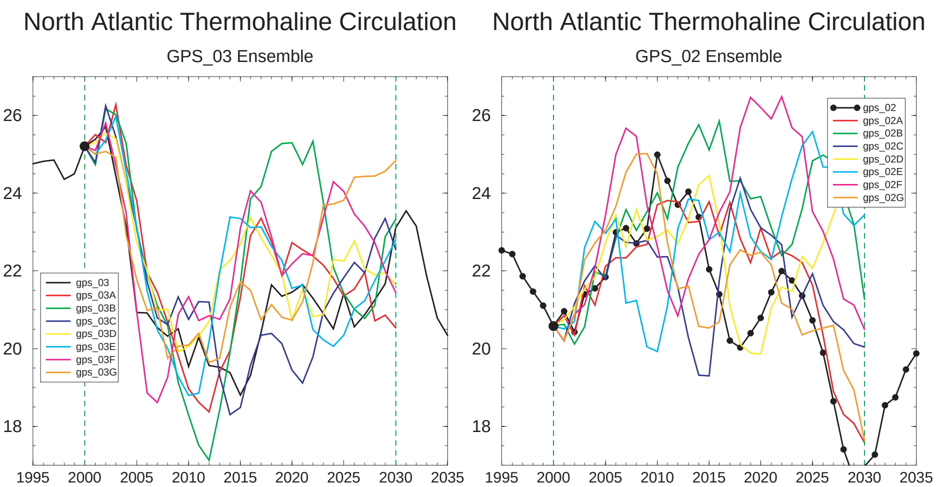

Figure 1: Evolution of the North Atlantic overturning circulation in two of the ensembles. (The third is qualitatively similar.) Within each ensemble the atmospheric weather differs at day one, but the ocean conditions are identical.

2 Experimental Design

The main experiments we describe are performed with a standard GFDL coupled climate model at R30 horizontal atmospheric resolution and 2°oceanic resolution (reference). A 900 year control integration with pre-industrial concentrations (in particular, from the year 1865) of forcing (greenhouse gases and aerosol concentrations) was performed to provide a stable climatology. Then, from three independent times in this integration anthropogenic greenhouse gases were increased for a period of 136 years, so providing three different initial conditions that each correspond to January, 2001. We may thus expect, for example, that the thermohaline conditions of the ocean differ for each state.

From each of these three states we then create and ensemble of eight members by perturbing the atmospheric initial conditions, and continuing the integrations for an additional 30 years, each with projected increasing levels of greenhouse gases from anthropogenic emissions. The experiments are similar to those of Griffies and Bryan (1997), except that the model has higher resolution in both atmosphere and ocean, and anthropogenic greenhouse gases and aerosols increase through the integration.

3 Results

Fig. 1 shows the divergence of the thermohaline circulation in the North Atlantic in two of the ensembles. Evidently, the model thermohaline circulation is predictable for a few decades. Furthermore, the natural variability of the thermohaline circulation over the next few decades dominates over the slow weakening

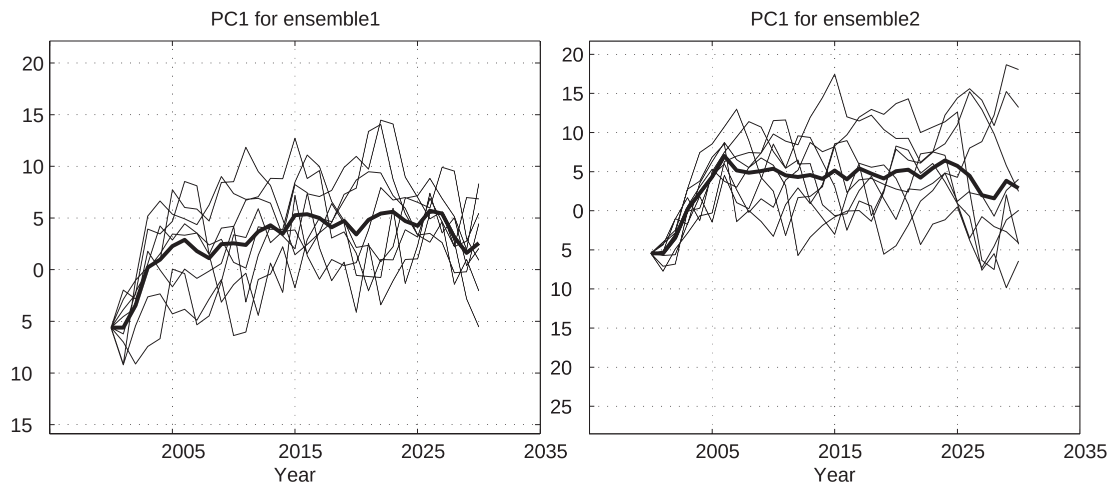

Figure 2: As for fig. 1,but for the first principal component of the two ensembles.

expected because of global warming. (The weakening arises because of increased precipitation at high latitudes, weakening the buoyancy forcing.)

The predictability of the SST is assessed using EOFs, calculated from a 300 year segment of the control integration. We restrict attention to the North Atlantic. Typically, the first few EOFs have a dominant signal in the subpolar gyre. The EOFs divergence of the principal components of the first EOFs for two of the ensembles is illustrated in fig. 2. Predictability of the SST seems present for a decade or perhaps two, after which the variance of any one integration is comparable to the difference between integrations.

4 Discussion and Conclusions

The thermohaline circulation in this model undergoes fairly regular oscillations with a dominant period of about 30, consistent with it being a damped oscillator driven by noise from the atmosphere (reference). This circulation is, perhaps not unexpectedly, predictable on this timescale. The surface manifestation of these oscillations and the associated predictability appears to be mainly in the subpolar gyre, consistent with theory that suggests that the thermohaline circulation is at first order buffered from the subtropical sea-surface by an isothermal layer at the base of the ventilated thermocline (e.g., Vallis 1999).

Our results suggest that predictability of the SST of the subtropical gyres may be lower than that of the subpolar gyres. This may be because, even though the subtropical gyres may also have internal decadal-scale variability, their mixed layer is shallower and the SST is determined to a greater degree (than in the subpolar gyre) by the intrinsically unpredictable atmospheric dynamics. To explore this we performed similar predictability experiments with a sector ocean model coupled to an atmospheric

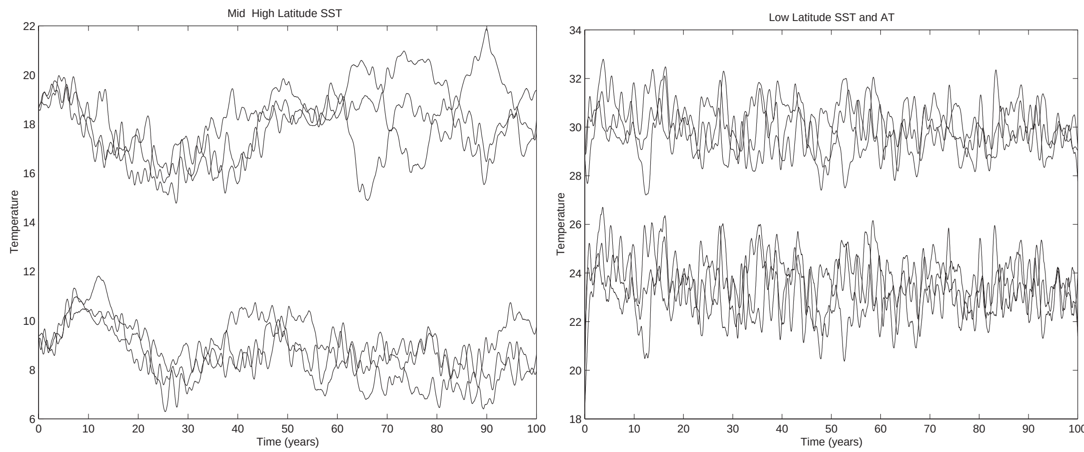

Figure 3: Evolution of a small ensemble of experiments of an a sector ocean model coupled to an atmospheric EBM. The ensemble members have different realisations of atmospheric noise. Left, SST at two latitudes in the subpolar gyre. Right, evolution of SST and SAT in the subtropical gyre.

energy balance model, plus noise whose variance is geographically uniform. Again the thermohaline circulation oscillates (in fact, sector models with flat bottoms typically oscillate more readily than models with realistic topography; Winton 19xx) providing high latitude SST predictability, but the subtropical SST (and SAT) is less predictable (fig. 3). The predictability properties of the lower latitudes, and the predictability of the atmosphere in general, remain topics for future exploration.

References

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